Haris Majeed Raja

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Summary

AI Engineer with hands-on experience building production-grade computer vision pipelines. Currently at MLBench, replacing multi-model client pipelines with fine-tuned RF-DETR and self-hosted language models, with a background spanning object detection, tracking, and full-stack AI application delivery.

Experience

AI Engineer Jun 2026 – Present
MLBench Pvt Ltd

- Worked on a field detection system for document forms, replacing a client's three-model pipeline (YOLO, FFDNet/FFDetr, and a geometry classifier) with a single fine-tuned **RF-DETR** model.
- Fine-tuned RF-DETR-Base on an annotated document dataset, reaching 0.9335 precision and 0.8496 recall against the client's existing three-model architecture at 0.9394 precision and 0.7875 recall, an F1 improvement from 0.8568 to 0.8896.
- Evaluated Tesseract and DocTR as OCR backends for the extraction pipeline, identifying checkbox glyph misreads and low-resolution footer text corruption in Tesseract output that DocTR resolved.
- Annotated client PDF documents for field geometry, type, and role to build the training set for the detection model, and logged experiment results to guide model selection.
- Contributed to a rice grain quality and variety analysis tool, training DINOv2-based classifiers across multiple image resolutions to grade grains as Full, Broken, Rejected, Weak, or Fatty and identify variety among Full grains.

Machine Learning Intern Jun 2025 – Aug 2025
Arch Technologies

- Developed multiple ML models including spam detection using TF-IDF and Naive Bayes, housing price prediction using Linear Regression, and digit and iris classification.
- Applied preprocessing, feature extraction, and evaluation metrics including accuracy, R^2, MAE, and MSE using scikit-learn, NLTK, and matplotlib.

Teaching Assistant – Software Engineering & Artificial Intelligence Aug 2025 – Jun 2026
Information Technology University

- Assisted in teaching core Software Engineering and Artificial Intelligence concepts, including system design, version control, testing methodologies, search algorithms, and fundamentals of machine learning.
- Mentored students on scalable software and AI projects, conducted code reviews, and graded assignments and exams with constructive feedback.

Projects

Multimodal Deepfake Detection using VLMs (FYP) Aug 2025 – Jun 2026
Supervised by Dr. Waqas Sultani

- Designed **ExDDV-Judge**, a three-stage detect, explain, judge pipeline replacing 11B/7B-parameter VLMs with lightweight alternatives, cutting model size by over 80% while running on a single consumer GPU.
- Solved a critical real-frame misclassification bug by building a **Dual-DB routing system** with a fine-tuned CLIP encoder, raising real-frame accuracy from 0% to 83%.
- Shipped a full-stack **Flask** web app delivering real-time deepfake analysis with per-frame labels, explanations, and trust scores.

Legal AI Chatbot | *GitHub Link* July 2026

- Built a full-stack RAG-based legal assistant using React, FastAPI, and MongoDB, with project-scoped retrieval, JWT-based authentication, and real-time token streaming powered by LangChain, FAISS, and the Groq API.

Footfall Counter | *GitHub Link* June 2026

- Built a real-time people-counting system using YOLO/RT-DETR and ByteTrack, with configurable virtual-line entry/exit counting, optimized to run entirely on CPU.

Technical Skills

AI/ML: PyTorch, HuggingFace, LangChain, FAISS, RAG, YOLO, RF-DETR, DINOv2, CLIP, ByteTrack, OpenCV, Tesseract, DocTR
Dev: Python, C++, Java, SQL, React, Node.js, FastAPI, Flask, MongoDB, scikit-learn, Gradio
Tools: Git, VS Code, Jupyter, Linux, JWT Auth